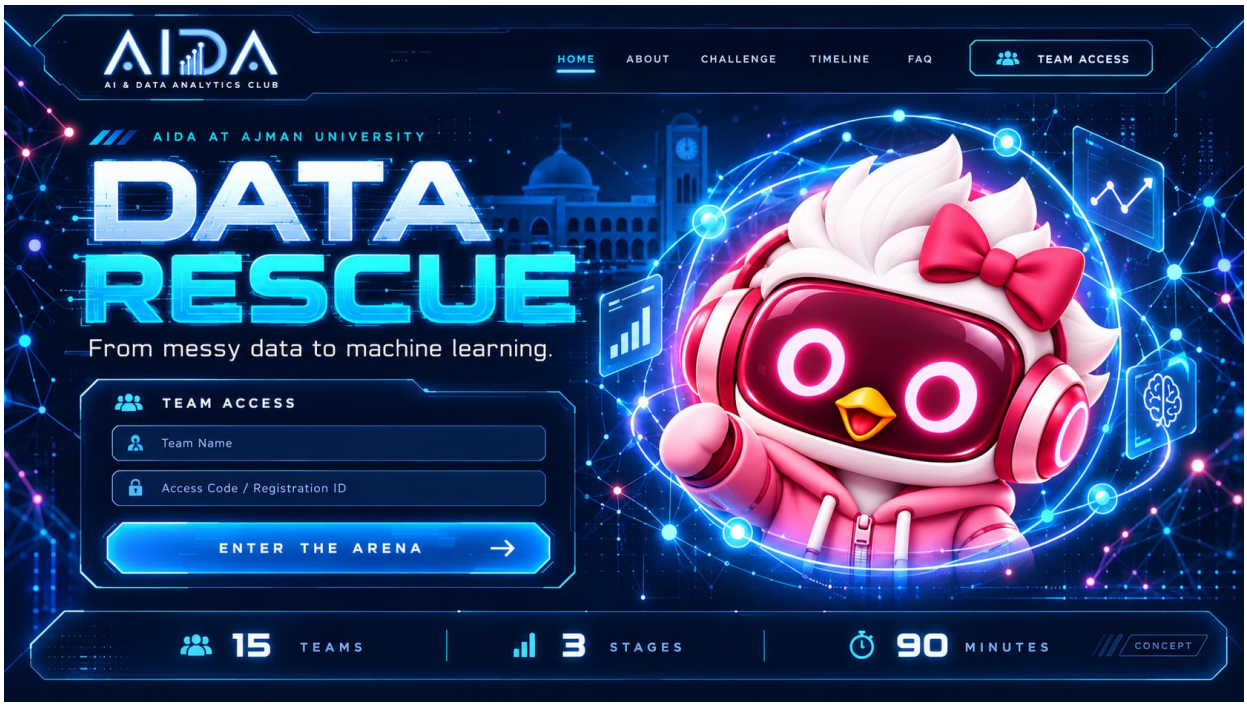
AIDA Data Rescue Hackathon Proposal

Ajman University

From Messy Data to Machine Learning



Concept image for the proposed participant experience

Decision requested	Proposed event
Approval	University lab, 45 students, sponsor outreach
Format	15 teams of three using one computer per team
Duration	90 minutes, including 60 minutes of competition
Funding ceiling	AED 4,000; sponsorship preferred
Preparation	Four weeks after approval

Prepared by the AI and Data Analytics Club

Executive Summary

The AI and Data Analytics Club requests approval to run AIDA Data Rescue, a 90-minute educational hackathon for 45 Ajman University students working in 15 teams. Participants will clean synthetic campus-equipment data, construct labels from maintenance records, and train a small neural network through three timed stages. The competition is designed for students with basic Python and pandas experience; starter code keeps the focus on data quality and model improvement.

We request a computer lab with 15 team computers and two spares, internet access, a projector, a faculty supervisor, two judges, one technical operator, and three student volunteers. We also request permission to approach sponsors through university-approved channels. The proposed spending ceiling is AED 4,000. No funding, sponsorship, venue, or infrastructure is currently assumed to be confirmed.

Approval Requested

- Approve the event concept, participant capacity, format, and four-week preparation period.
- Reserve a suitable university computer lab and technical support for a rehearsal and the event.
- Authorize institutional sponsor outreach led by a faculty adviser, Student Life, or the Ajman University Innovation Center.
- Approve a spending ceiling of AED 4,000, subject to university purchasing and sponsorship rules.

Expected Learning Outcomes

- Recognize missing values, malformed numeric data, duplicates, impossible readings, and identifier inconsistencies.
- Create reliable labels by normalizing and joining a separate outcome log.
- Prevent data leakage by fitting preprocessing only on the training partition.
- Use validation accuracy and binary cross-entropy loss to compare neural-network experiments.
- Explain technical decisions and work responsibly with limited AI assistance.

Event Participants and Staff

Role	Count	Main responsibility
ents	45	Compete in 15 teams of three
lty supervisor	1	Academic oversight and final decisions
es	2	Review evidence and approve scores
nical operator	1	Manage clock, platform, incidents, and judging
ent volunteers	3	Registration, room support, and participant questions

Competition Format

The official competition lasts 60 minutes. Three stages run for 20 minutes each, followed by 30 minutes for final scoring, provisional results, awards, and photographs. Registration, login checks, the guide, and environment checks happen before the official clock starts.

Time	Activity	AI access
00:00 to 20:00	Stage 1 Rescue the Data	Unlocks at 05:00
20:00 to 40:00	Stage 2 Build the Labels	Unlocks at 25:00
40:00 to 60:00	Stage 3 Train and Improve	Unlocks at 45:00
60:00 to 70:00	Final scoring and verification	Closed
70:00 to 80:00	Publish provisional results	Closed
80:00 to 90:00	Awards and photographs	Closed

Stage 1 Rescue the Data

Students normalize identifiers and numerical values, remove duplicate copies, convert impossible readings to missing values, and document an imputation strategy. Learned imputation values must come from the fixed training partition.

Criterion	Points
ma, parsing, units, and ranges	10
plication and identifier preservation	10
ng-value handling and data-quality report	10

Stage 2 Build the Labels

Students clean and join a separate maintenance-outcome log. Corrective repair confirmed maps to 1; inspection complete with no corrective repair maps to 0. Unmatched records must be reported. Outcome text, record IDs, and target values cannot be model features.

Criterion	Points
ct labels matched to records	15
split and target-leakage prevention	10
le numerical input and consistent preprocessing	5

Stage 3 Train and Improve

Students begin with a 12 input, 16 hidden, 8 hidden, and 1 output PyTorch network. They may tune the learning rate, batch size, dropout, weight decay, hidden widths, and epoch count up to 50. Each training run is limited to 120 seconds on identical CPU resources.

Criterion	Points
te-test accuracy	20
te-test binary cross-entropy loss	10
oducible model and preprocessing	5
nation of two experiments	5

Dataset Design

Campus Equipment Rescue is a fictional dataset created for education. It contains no student information, personal data, or confidential university records. The task is to predict whether a piece of equipment requires corrective maintenance within seven days.

Component	Specification
Unique records	10,000
Participant development file	8,200 rows: 8,000 unique records plus 200 duplicate copies
Training and validation	6,000 training and 2,000 validation records
Private test set	2,000 records held by organizers
Input variables	12 numeric equipment readings
Target	needs_maintenance, created by students
Outcome log	8,000 development records
Observed positive rate	49.05 percent

Input Variables

Variable	Meaning
temperature_c	Equipment temperature
ambient_temperature_c	Ambient temperature
vibration_mm_s	Vibration velocity
current_a	Electrical current
voltage_v	Electrical voltage
humidity_pct	Relative humidity
pressure_bar	Operating pressure
rotation_rpm	Rotation speed
daily_runtime_hours	Daily operating time
equipment_age_months	Equipment age
days_since_service	Time since service
load_pct	Operating load

Controlled Data Problems

- Approximately 5 percent of feature cells are missing.
- Some numeric values contain units or extra spaces.
- A small number of readings exceed published physical ranges.
- Two hundred duplicate copies are mixed into the development file.
- Record identifiers and maintenance statuses use inconsistent spacing and capitalization.

The organizers retain a clean reference dataset, private test labels, and a corruption manifest. Participant downloads exclude those files.

Scoring and Fairness

The judges compute scores from frozen submissions and actual predictions. Participants cannot score their own private-test results. A fresh isolated run must reproduce the submitted artifacts before the model receives reproducibility points.

Metric Formula

Accuracy points = 20 x clamp((accuracy - 0.50) / 0.40, 0, 1). Loss points = 10 x clamp((ln(2) - loss) / (ln(2) - 0.30), 0, 1). Accuracy is expressed as a fraction. The clamp function limits each normalized value to the interval from zero to one.

The reference implementation completed in approximately 4.4 seconds on a local CPU. It reached 88.8 percent validation accuracy, validation loss of 0.273, private-test accuracy of 87.7 percent, and private-test loss of 0.281. These results support a practice target of at least 85 percent validation accuracy and validation loss no greater than 0.40.

Tie Resolution

Ties are resolved by private-test accuracy, then private-test loss, then the Stage 1 score. If teams remain tied, they share the placing and the related prize pool is divided equally.

Submission Rules

- The server clock controls all deadlines and stage unlocks.
- Teams may replace a submission before the deadline; the latest accepted receipt counts.
- Autosave preserves work but does not count as submission.
- At the deadline, the last explicitly submitted version is frozen.
- A missed stage earns zero for that stage, but the team may continue.
- Early submission does not unlock the next stage.
- Previous-stage code remains visible as a read-only snapshot.

Manual Review

Automated checks verify row coverage, schema, ranges, labels, preprocessing metadata, model execution, and private-test metrics. Faculty judges review the imputation rationale and the explanation of two experiments. No leaderboard is published until the judges approve the evidence and the technical operator records a reason for every score.

Limited AI Assistance

Each team receives one built-in AI prompt in each stage, for a maximum of three prompts per team and 45 successful responses across the event. The assistant supports learning by explaining a concept, interpreting a short error, or suggesting an experiment. It cannot access the dataset, run code, edit files, or complete the full pipeline.

Rule	Limit
Unlock time	Five minutes after each stage begins
Prompt size	80 whitespace-separated words and 1,000 characters
Response size	160 displayed words
Illustrative code	No more than five lines
Follow-up conversation	Not available
Unused prompts	Do not carry forward
Model settings	Same organizer-selected model for all teams

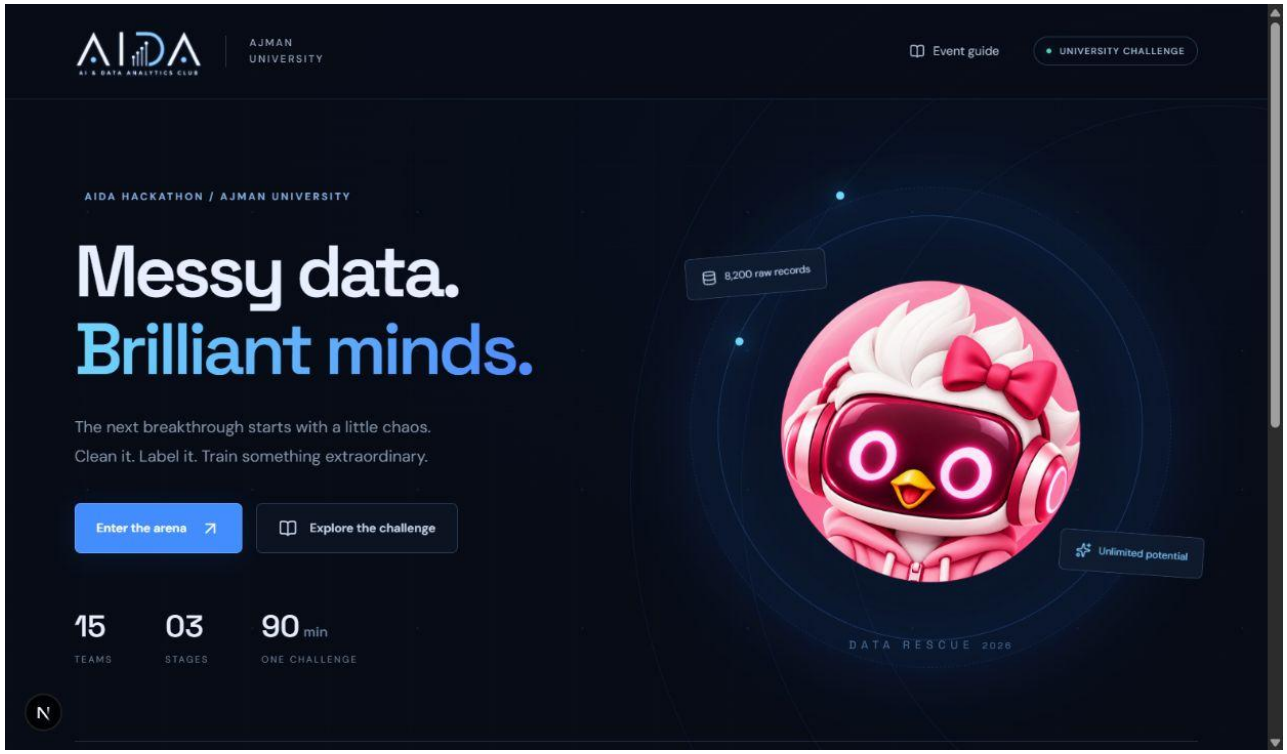
Control Measures

- The server atomically claims one prompt for each team and stage.
- Duplicate clicks return the same claimed request state.
- A provider failure allows the same prompt to be retried; it does not create a second prompt.
- The system instruction restricts the assistant to conceptual help and a short response.
- Only the built-in assistant is permitted on supervised university computers.
- If AI fails for every team, the organizer pauses the shared event clock and logs the incident.

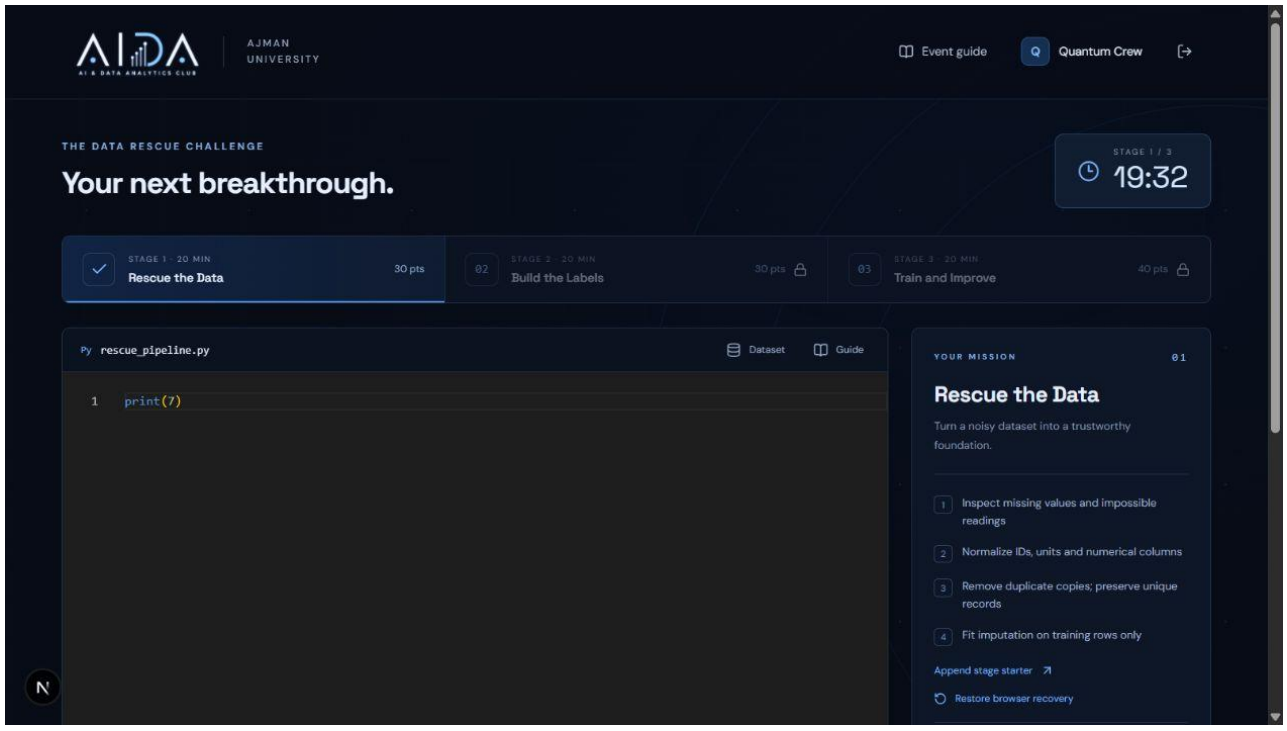
Prompt and response limits reduce the chance that the assistant completes the work, but they cannot guarantee this. The scoring rubric therefore rewards reproducibility and a short explanation of the team's own experiments.

Participant Platform

The implemented desktop-first website includes team login, dataset preview and downloads, a guide sheet, a Python editor, console output, training metrics, a server-synchronized timer, AI prompt controls, autosave status, submission receipts, prior-stage snapshots, and organizer controls.



Implemented local website landing page



Implemented participant workspace

Platform Architecture

Visual Design

The interface uses the supplied AIDA logo, deep navy surfaces, electric blue borders, cyan highlights, connected-node patterns, and the supplied mascot without recoloring it. Motion is disabled when the operating system requests reduced motion. The one-minute warning does not interrupt typing.

Subsystem	Implementation
Web application	Next.js and TypeScript on Vercel
Code editor	Monaco Editor, packaged locally
Identity and durable state	Supabase Auth and Postgres
Submission artifacts	Private Supabase Storage bucket
Python execution	One isolated Vercel Sandbox per active team
Machine learning	Pinned CPU PyTorch environment
AI help	OpenAI Responses API, server-side only
Judging	Fresh isolated run followed by trusted inference and scoring

Security and Integrity

- Participant code runs in a remote microVM with outbound network access denied.
- Application secrets and private test labels are never mounted into participant environments.
- Team requests are authenticated on the server; one team cannot read another team's work.
- Supabase row-level security is enabled and direct browser access to event state and artifacts is revoked.
- The OpenAI API key, Supabase service credential, and Vercel credential remain server-side.
- The judging system supplies private test features to trusted inference only; scoring labels remain outside the execution sandbox.
- A 120-second command timeout, two CPU threads, file allowlists, and 10 MB artifact limits constrain runs.
- Audit records capture event-control actions and submission receipts.

Hosting Requirements

Vercel's published Hobby allowance permits 10 concurrent sandboxes, which is below the 15-team requirement. The event therefore needs a paid Vercel allocation or approved university infrastructure. Supabase currently lists a free plan for rehearsal and a Pro plan starting at USD 25 per month. Final procurement must verify current prices and university rules before purchase.

Deployment Status

The application, dataset generator, reference model, scoring scripts, and deployment configuration are complete locally. A cloud deployment is not active because the connected Supabase account contains no project and no Vercel project credentials were supplied. The project fails closed when those services are absent; participant code is never executed on the organizer's computer.

Budget Request

The figures below are planning allowances rather than supplier quotations. No expenditure is currently approved. University facilities, equipment, staff time, and platform development are requested in kind.

Item	Calculation	AED
First-place prize	AED 400 x 3 students	1,200
Second-place prize	AED 250 x 3 students	750
Third-place prize	AED 150 x 3 students	450
Refreshments	55 people x AED 12	660
Trophies or medals	Allowance	150
Printing and signs	Allowance	100
Hosting and execution	Event and rehearsal allowance	200
LLM API usage	Event and testing allowance	50
Subtotal		3,560
Contingency	10 percent	356
Estimated total		3,916
Requested ceiling	Rounded	4,000

Lower Cost Option

A technical-only version can target AED 300 by using university facilities, digital certificates, no catering, and donated prizes. A zero-cash event requires confirmed in-kind compute, AI access, and prizes before registration opens.

Prize Conditions

Monetary prizes should not be advertised until funding is secured. Product or voucher prizes should provide equal value to all three students on a winning team. Shared placings divide the relevant prize pool equally.

In Kind Requests

- Computer lab with 15 team computers and two spares.
- Stable internet connection and a projector.
- Technical rehearsal access before the event.
- Faculty supervisor, two judges, one technical operator, and three volunteers.
- Institutional sponsor introductions and purchasing support.

Sponsor Approach

Sponsor outreach should start with university relationships. A faculty adviser, Student Life, or the Ajman University Innovation Center should make or approve introductions. The club should present the event as a pilot and avoid offering a long-term official sponsor title without university approval.

Organization	Proposed request	Contact route
Ajman University Innovation Center	Introductions, mentors, judges, or infrastructure	auic@ajman.ac.ae; +971 6 705 6481
Ajman Bank	Approximately AED 2,400 for prizes	University partnership owner; then info@ajmanbank.ae
Mbank	Prizes, guest judge, or educational vouchers	University office managing the existing AU-Mbank relationship
AUIC startups and suppliers	Refreshments, printing, or technology gifts	Introduction through AUIC or Student Life

Suggested Packages

Package	Request	Recognition subject to approval
Lead sponsor	AED 2,400	Approved logo placement, acknowledgement, and award invitation
Supporting sponsor	AED 1,000	Acknowledgement and event invitation
Technology partner	AED 300 to 600 or in kind	Acknowledgement for compute, API credits, or vouchers

Draft Sponsor Message

Subject: Sponsorship request for AIDA Data Rescue at Ajman University

Dear [Name],

The AI and Data Analytics Club at Ajman University is seeking support for AIDA Data Rescue, a 90-minute educational hackathon for 45 students. Fifteen teams will clean a deliberately messy dataset, construct labels from evidence, and train a neural network in a supervised university lab.

We invite [Organization] to support student prizes, refreshments, technology costs, or a guest judge. The proposed lead contribution is AED 2,400, which funds all three winning teams. Any recognition, logo placement, or public announcement will follow Ajman University approval.

We would welcome a short meeting to share the scoring rubric, technical controls, schedule, and expected student outcomes. Thank you for considering this request.

Sincerely,
AI and Data Analytics Club
Ajman University

Preparation Schedule

Week	Work	Completion evidence
1	Obtain approval; confirm venue and faculty roles; freeze dataset and reference solution	Signed approval path, room request, validated dataset
2	Configure cloud services; test login, timer, submissions, and AI quotas	Deployment checklist and service test results
3	Add final branding; prepare volunteers and judging workflow; contact sponsors	Rehearsal script, judge guide, sponsor status
4	Run a 15-team rehearsal; calibrate targets; freeze rules and funding	Signed readiness checklist and incident plan

Readiness Checklist

- Fifteen teams can execute code concurrently with identical resources.
- Beginner pilot teams can complete each stage within 20 minutes.
- The baseline model trains within two minutes.
- Cleaning improves data quality without target leakage.
- Judges reproduce submissions and private-test scores.
- Teams cannot read private labels or another team’s files.
- AI unlocks and quotas survive duplicate requests, refreshes, and provider failures.
- Deadlines reject late submissions and preserve accepted versions.
- All submissions are judged within the first ten minutes after competition.
- Animations do not interfere with editing or accessibility.
- Spending alerts, environment shutdown, and the outage procedure have been rehearsed.

Acceptance Results Completed Locally

Check	Result
TypeScript and production build	Passed
Timing, prompt, and scoring unit tests	5 of 5 passed
Authentication, team isolation, deadlines, and data-gating API checks	10 of 10 passed
Dependency security audit	0 known vulnerabilities after pinning
Reference model private-test accuracy	87.7 percent
Reference model private-test loss	0.281
Cloud concurrency rehearsal	Pending paid Vercel and Supabase configuration

Risks and Controls

Risk	Control
Cloud service interruption	Organizer pauses the shared clock, records the incident, and applies the same adjustment to every team.
Late or lost work	Five-second autosave, browser recovery copy, durable remote revisions, and explicit submission receipts.
AI assistant completes too much work	Short response, five-line code maximum, no tools, no dataset access, one prompt, and explanation points.
Data leakage	Fixed split, training-only preprocessing checks, private labels outside sandboxes, and trusted inference.
Unequal computing resources	One remote two-vCPU sandbox per team with the same image and timeout.
Sponsor funding is delayed	Do not advertise cash prizes; use the AED 300 technical-only option or postpone registration.
Student code is unsafe	Isolated microVMs, denied outbound network, no application secrets, timeouts, and artifact limits.
Scores are disputed	Frozen receipts, rerunnable code, audit records, published formulas, faculty review, and a ten-minute question period.

Decision and Signatures

Approval permits the club to begin the four-week preparation schedule and institutionally approved sponsor outreach. Event publication and prize advertising remain conditional on confirmed venue, technical capacity, judging staff, and funding.

Decision	Name and signature	Date
Approved / Not approved		
Dean or authorized representative		
Faculty supervisor		
Student Life representative		
Technical readiness confirmation		

Administrative Details

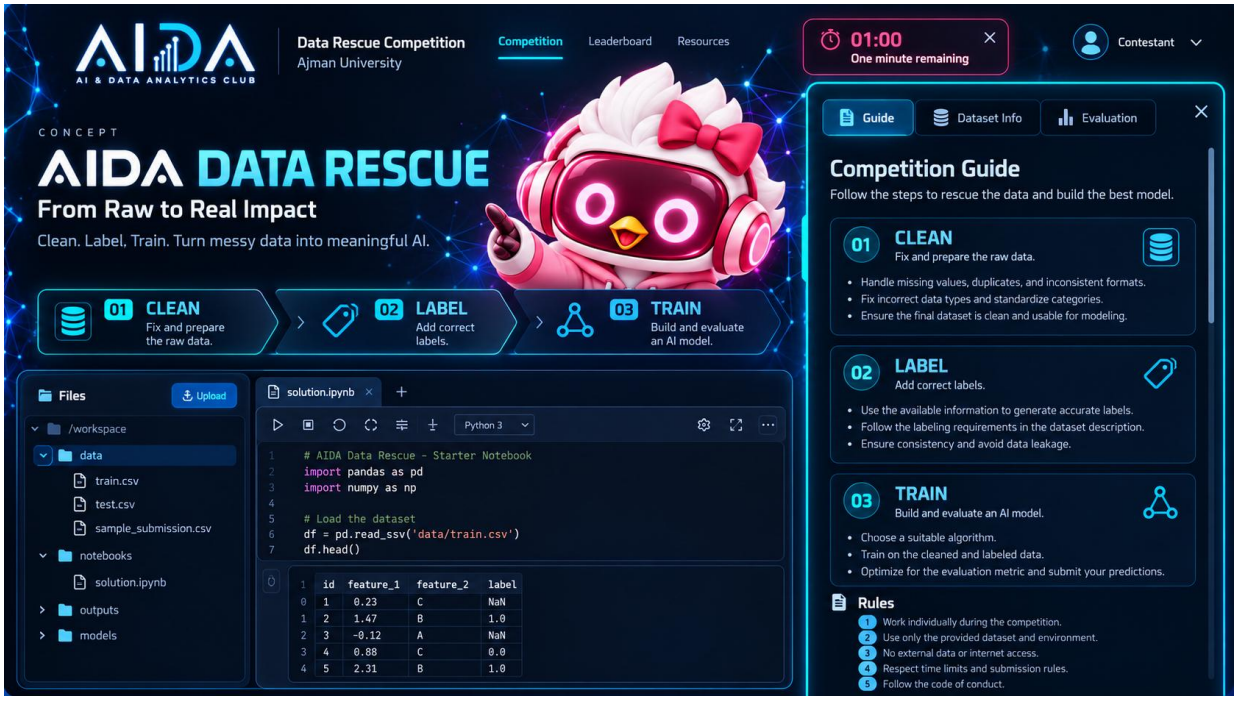
Field	To be completed
Event date	
College	
Venue	
Registration opening	
Registration closing	
Faculty supervisor	
Technical operator	

Visual Concepts

These generated images show the intended visual direction. They are concepts rather than screenshots of the working application. The preceding platform section contains screenshots of the implemented local website.



Concept for the participant workspace



Concept for the guide and time warning



Concept for the transition and results experience

Sources

Ajman University Innovation Center
https://auic.ajman.ac.ae/
Contact information and partnership role.

Ajman University and Ajman Bank
https://www.ajman.ac.ae/en/mou/ajman-bank
Existing university relationship.

Ajman Bank customer service charter
https://www.ajmanbank.ae/site/customer-service-charter.html
General inquiry contact route.

Ajman University and Mbank
https://www.ajman.ac.ae/en/news/2024/mbank-develops-strategic-partnership-with-ajman-university-to-offer-academic-and-professional-development-programs-to-university-students
Existing partnership concerning student development.

Vercel Sandbox pricing and quotas
https://vercel.com/docs/sandbox/pricing
Current concurrency, duration, and usage information; verify before purchase.

Supabase pricing
https://supabase.com/pricing
Current plan information; verify before purchase.

OpenAI Responses API
https://developers.openai.com/api/reference/resources/responses/methods/create
Server-side AI response configuration.

Pricing and quotas can change. The organizing team must verify current provider terms, university procurement rules, sponsor approval, and data-protection requirements before deployment or purchase.